

Local DOCX PII Redactor

Plain-English user guide for reviewing and redacting Word documents on your Mac

Privacy promise

The utility runs on your Mac. It does not use a cloud API, does not use an LLM, and does not change your original Word file.

What This Tool Is For

Use this local app when you have a long Word document, such as Teams or email transcripts, and need help finding possible names, email addresses, phone numbers, and identifying numbers before you manually decide what to redact.

- It scans the DOCX and groups repeated values so you do not have to read every page first.
- It shows context snippets so you can decide whether each value should stay or be redacted.
- It creates a new redacted DOCX plus an audit log and reusable decisions file.
- It treats detection as imperfect. You stay in control of every redaction decision.

Quick Start

From Terminal on your Mac:

Open folder	<code>cd /Users/agoodloe/Documents/Codex/2026-07-24/before-making-any-implementation-decisions-read/work/pii_docx_redactor</code>
Start app	<code>python3 -m streamlit run app.py --server.headless true --server.port 8501 --browser.gatherUsageStats false</code>
Open browser	Go to http://localhost:8501

The app may already be running. If the page opens, you can start using it immediately.

Using the App

1. Upload a DOCX file. Keep the original file somewhere safe; the app creates a separate redacted copy.
2. Click Scan document. The scan runs locally and may take a little while for a long document.
3. Review each tab: People, Numbers/IDs, Email addresses, Phone numbers, and Other possible identifiers.
4. For each candidate, choose Keep everywhere, Redact everywhere, Review each occurrence, or leave it Undecided.
5. Use the context snippets to understand where the value appears without reading the whole document.
6. If you choose Review each occurrence, decide Keep or Redact for each listed occurrence.
7. When the safety check looks right, generate the output files.

Decision Choices

Keep everywhere	The value remains unchanged in the redacted output.
Redact everywhere	Every matching occurrence is replaced unless an occurrence-specific rule says otherwise.
Review each occurrence	You decide occurrence by occurrence. This is useful when the same word is sometimes a name and sometimes not.
Undecided	A holding state while you review. The app warns you before output generation.

Replacement Labels

People default to consistent pseudonyms such as [PERSON 001], [PERSON 002], and so on. You can change a replacement label before generating output.

- Example: Jane Smith -> [STUDENT 01]

- Example: robert.lee@example.edu -> [REDACTED EMAIL]
- Example: 123456789 -> [REDACTED ID]

Files You Get Back

Redacted DOCX	<original_name>_redacted.docx
Audit log	<original_name>_redaction_log.csv
Reusable decisions	<original_name>_decisions.json

The CSV log records the original candidate, replacement, type, detector source, decision, occurrence count, context, timestamp, and the SHA-256 hash of the input file.

What the Scan Looks At

- Body paragraphs
- Tables, including nested tables where accessible
- Headers and footers
- Hyperlink display text when Word exposes it as normal text
- Core document metadata, listed separately for review

Important Limits

Final review required

Do not treat the output as automatically safe. Open the redacted DOCX in Word and do a final human review before sharing it.

- Name detection is helpful but imperfect, especially in messy transcripts.
- The fallback detector handles First Last and Last, First patterns, but it may miss informal names or unusual capitalization.
- Footnotes, endnotes, comments, text boxes, SmartArt, embedded objects, tracked changes, and unusual Word structures may not be redacted perfectly in this MVP.

- The tool tries to avoid dates and page numbers when looking for numeric IDs, but you should still review number candidates carefully.

Optional Better Name Detection

If you have Python 3.11+ and want stronger local person-name detection, you can optionally install spaCy and its small English model. This still runs locally.

Install spaCy	<code>pip install spacy</code>
Install model	<code>python -m spacy download en_core_web_sm</code>

If spaCy is not installed, the app still works using deterministic local patterns.

Testing With Fake Data

To make a synthetic DOCX with invented names and identifiers:

Generate sample	<code>python sample_docx_generator.py</code>
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The sample file is safe test data and does not contain real student information.

Good Working Habit

1. Scan the original DOCX.
2. Review and decide candidates.
3. Generate the redacted DOCX and audit files.
4. Open the redacted DOCX in Word.
5. Search the redacted output for any especially sensitive original names or numbers you know about.
6. Keep the CSV and JSON beside the redacted file for your record.